NABIN KUMAR KC

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Education

Wichita State University **Jan 2025 – Dec 2026 (Expected)**
Master of Science in Computer Science *Wichita, Kansas*

Nitte Meenakshi Institute of Technology **Aug 2019 – July 2023**
Bachelor of Engineering in Computer Science Engineering *Bangalore, India*

Technical Skills

Programming: Python, SQL, C, JavaScript, HTML/CSS
Machine Learning: Scikit-learn, Random Forest, KNN, Logistic Regression, Linear Regression
Deep Learning: TensorFlow/Keras, CNNs, Image Classification, Data Augmentation
Computer Vision: OpenCV, MediaPipe, Image Preprocessing, Feature Detection
Data/Analytics: Excel, Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Preprocessing, Feature Engineering
Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab, Linux, Vercel
Cybersecurity: Wireshark, NFStream, Network Traffic Analysis

Experience

Wichita State University **Jan 2026 – Present**
Graduate Research Assistant (Part-time) *Wichita, Kansas*
- Developed an OCR admissions document pipeline (OpenCV, regex, fuzzy matching) achieving 91% field-level accuracy.
- Evaluated OCR robustness on 360 simulated/scanned documents across multiple image-quality conditions.
- Built and evaluated Decision Tree, Random Forest, and Logistic Regression models for applicant decision support.
- Conducted benchmarking experiments and presented model performance and recommendations to faculty.

YBI Foundation **Sep 2022 – Nov 2022**
Machine Learning with Python Intern *Remote*
- Performed exploratory data analysis and visualization with Pandas, Matplotlib, and Seaborn.
- Preprocessed data by handling missing values, encoding categorical features, scaling, and applying PCA.
- Built regression and classification models (KNN, Random Forest), evaluated with R^2, accuracy, and confusion matrices.

The Sparks Foundation **Jun 2022 – Aug 2022**
Frontend Web Developer Intern *Remote*
- Designed and developed a multi-page donation website (Home, Donate, Contact) using HTML, CSS, and JavaScript.
- Integrated the Razorpay payment gateway to enable secure online donations.
- Implemented interactive animations using JavaScript and GSAP, along with a contact form to improve user engagement.

Projects

OCR-Based Admissions Document Extraction | *Python, OpenCV, OCR, Scikit-Learn* **Jan 2026 – Present**
- Built a 4-stage OCR pipeline (OpenCV preprocessing, adaptive thresholding, regex extraction, fuzzy matching).
- Developed an automated Admit/Review/Deny workflow combining OCR-extracted features with ML classifiers.

Hand Gesture Control System *(Published, Springer 2024)* | *Python, OpenCV, MediaPipe, Tkinter* **Jun 2023**
- Built a desktop app to control a computer with webcam hand gestures, using OpenCV and MediaPipe tracking.
- Mapped gestures to volume, brightness, mouse, and media controls via a Tkinter GUI packaged as an executable.

Malicious DoH Traffic Detection | *Python, TensorFlow, Scikit-Learn, NFStream*
- Detected malicious DNS-over-HTTPS (DoH) tunneling traffic using network flow features extracted with NFStream.
- Compared logistic regression, random forest, and CNN models; tested robustness to adversarial evasion.

Plant Disease Detection System | *Python, TensorFlow/Keras, CNN, OpenCV* **Dec 2025**
- Built a custom CNN for plant disease detection, with augmentation and normalization for robustness.
- Deployed the model in a web app where users upload leaf images for real-time disease predictions.

Fingerprint Spoof Detection using CNN | *Python, TensorFlow/Keras, OpenCV*
- Preprocessed images with OpenCV (histogram equalization, thresholding, Canny edges) and augmentation.
- Trained CNNs to classify live vs. spoofed fingerprints, optimized via hyperparameter tuning.

Certificates

Machine Learning | Google Cybersecurity | Data Analysis with Python